From Transparent to Opaque: Leveraging GRU in Latent Reasoning Chain

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Abstract

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1 Introduction

The Chain of Continuous Thought (COCONUT) [Hao et al., 2024] explores the potential of the latent layer of reasoning, as an alternate to traditional explicit CoT [Wei et al., 2022] in both reasoning procedure or answering procedure. Its prior work has shown that the latent layer of continuous reasoning can reduce unnecessary statements, and focus on the core of the problem.

These model issued are trained from general-purpose language, being supervised to train from steps of reasoning procedure, to progressively increase its layers of latent reasoning. It's impractical due to lack of customized reasoning data, and the progressive training is significantly less convenient to explicit CoT. Hence, we provide a new approach to implement these latent reasoning layer—fine-tuning from explicit reasoning models.

In 2014 [Chung et al., 2014], Chung et al. proposed the Gated Recurrent Unit. It is firstly leveraged for improving the RNNs, but we found that it is also feasible to be used in latent reasoning chain, to maintain its long-term contextual awareness.

In January, the DeepSeek team proposed a distillation-based method for implementing a reasoning model via reinforcement learning [DeepSeek-AI, 2025], and they applied the approach to the Qwen-2.5B [Yang et al., 2024] to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement the COCONUT on the small model, and find its potential in reasoning, compared to the original explicit reasoning model.

2 Methodology

2.1 Architecture

We believe that training a language model from direct answering to latent reasoning is impractical without specialized guidance. Instead, we propose a direct method, alternating the model architecture from purely connection of latent reasoning layers, to a GRU-based latent reasoning chain.

We deploy a reset gate and an update gate, not during the model architecture, but the reasoning process.

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2.1.1 Language Model

We use the proposed architecture proposed in COCONUT [Hao et al., 2024]. It enables two modes for LLMs—latent mode and the transparent mode.

Latent Mode The latent mode is designed to reduce extra computation of decoding reasoning text. Hao et al. demonstrates that human language is designed to communicate instead of heavily on reasoning, so we believe it's unnecessary for users to always decipher what AI reasons. This mode also exclusively own the gated unit, which will be explained later.

Transparent Mode This is the normal mode of what LLMs do. They speak human language, and eventually outputs answer during our research progress.

2.1.2 Gated Unit

The gated unit is the innovative point of our paper.

Unlike standard causal models that treat each decoding step as conditionally independent beyond the token history, we introduce a GRU-style gating mechanism in the latent reasoning stage. This gate learns to control information flow and retention, enabling the model to persist and refine internal reasoning states across steps. Effectively, this transforms reasoning from a fully recurrent process into a gated latent update, yielding better consistency and longer-term dependency tracking.

2.2 Reinforcement Learning

Inspired by DAPO [Yu et al., 2025], we deployed a mechanism that applies standardization considered beyond accuracy.

2.2.1 Sampling

They key to let model learn is to sample, and filter satisfying result.

The full sampling procedure is shown in figure 2. We will provide our hyperparameters in appendix.

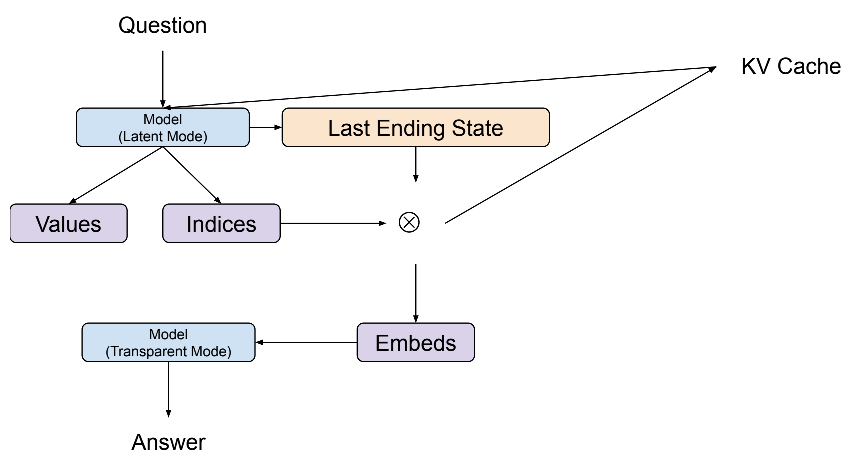

Figure 1: The sampling procedure. It happends with parallelized compuation, but for the sake of simplicity, we just use one as example.

2.2.2 Award Mechanism

That's because we're aware that DeepSeek's method unintentionally increased the reasoning procedure into an abnormal length.

Since its batched, the final formula is given on 1, where r_i, c_i, and l_i denotes to award, correctness award, and length award respectively.

$$r_i = \frac{c_i - \bar{c}}{\sqrt{\sum (c_i - \bar{c})^2} + 10^{-6}} + \frac{l_i - \bar{l}}{\left|\max\left\{l_i - \bar{l}\right\}\right| + 10^{-6}} \tag{1}$$

By adding 10^{-6}, we can avoid producing `NaN` if dividing zero. Then we apply these scores to the loss, given by 2. We are also leveraging gradient accumulation during the training procedure.

$$\text{loss} = \sum \text{model loss} + \frac{\overline{r_i}}{\sum \text{text end indices} + 1} \tag{2}$$

And to promote its diversity, we are also applying punishment for being identical.

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